

Sampling

- ❑ Sampling: obtaining a small sample s to represent the whole data set N
- ❑ Allow a mining algorithm to run in complexity that is potentially sub-linear to the size of the data
- ❑ Key principle: Choose a **representative** subset of the data
 - ❑ Simple random sampling may have very poor performance in the presence of skew
 - ❑ Develop adaptive sampling methods, e.g., stratified sampling.
- ❑ Note: Sampling may not reduce database I/Os (page at a time)

Types of Sampling

- ❑ **Simple random sampling**

- ❑ Equal probability of selecting any item

- ❑ **Sampling without replacement**

- ❑ Once an object is selected, it is removed from the population

- ❑ **Sampling with replacement**

- ❑ A selected object is not removed from the population

- ❑ **Stratified sampling**

- ❑ Partition (or cluster) the data set, and draw samples from each partition (proportionally, i.e., approximately the same percentage of the data)

